Analysis tutorial 2

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1 Introduction

Why do we need Analysis?

We begin by showing few *fake proofs* that illustrate paradoxical results, which careful analysis can resolve.

Example 1

Consider the series

$$S = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots .$$

A naive manipulation gives

$$2S = S + 1 \quad \Longrightarrow \quad S = 1.$$

Now, consider instead

$$S = 1 + 2 + 4 + 8 + \cdots .$$

Then

$$2S + 1 = S \quad \Longrightarrow \quad S = -1.$$

This illustrates how careless handling of infinite series can lead to contradictions.

Example 2

For finite sums, we have

$$\sum_{i=1}^{m}\sum_{j=1}^{n} a_{ij} = \sum_{j=1}^{n}\sum_{i=1}^{m} a_{ij}.$$

But for infinite sums, this is not always valid:

$$\sum_{i=1}^{\infty}\sum_{j=1}^{\infty} a_{ij} \neq \sum_{j=1}^{\infty}\sum_{i=1}^{\infty} a_{ij}.$$

Consider the infinite matrix

$$\begin{pmatrix} 1 & 0 & 0 & 0 & \cdots \\ -1 & 1 & 0 & 0 & \cdots \\ 0 & -1 & 1 & 0 & \cdots \\ 0 & 0 & -1 & 1 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

Question: Where did we make mistakes?

2 Peano Axioms and Induction

The Peano axioms form the foundation of the natural numbers. A central feature is the **principle of induction**, which can be outlined as follows:

$$\big(P(0) \text{ holds, and } P(n) \Rightarrow P(n+1)\big) \quad \Longrightarrow \quad P(n) \text{ holds for all } n \in N.$$

Induction is a brilliant idea that is used to solve numerous problems in mathematics. There are different variations: Standard induction, Strong induction, Backwards induction, and the idea can be generalized for different situations as well.

Strong Induction looks like this:

Proof: We will show that $P(n)$ is true for all n, using induction on n.

Base case: Show that $P(1)$ is true.

Induction step: Suppose that

$$P(1),\ P(2),\ \ldots,\ P(k)$$

are all true for some integer k. We need to show that

$$P(k+1)$$

is true.

Exercises:

Problem 1. Show that

$$1 + 2 + 2^2 + \cdots + 2^n = 2^{n+1} - 1, \quad n \in N.$$

Problem 2. Prove that

$$\sum_{i=1}^{n} i \cdot 2^i = (n-1) \cdot 2^{n+1} + 2.$$

Problem 3. Show that

$$\prod_{k=2}^{n} \left(1 - \frac{1}{k^2}\right) = \frac{n+1}{2n}.$$

Problem 4. Prove that if $\alpha > 2$, then

$$\alpha^n > \alpha^{n-1} + \alpha^{n-2} + \cdots + \alpha + 1.$$

Problem 5. Consider the sequence of real numbers $(x_n)_{n\geq 1}$ defined by

$$x_1 = 1, \qquad x_{n+1} = \sqrt{1 + 2x_n} \quad \text{for } n \geq 1.$$

Use the Principle of Mathematical Induction to show that $x_n < 4$ for all $n \geq 1$.

Problem 6. Show that there are infinitely many natural numbers n such that

$$n! > 3^n.$$

Problem 7. In the parlour game *Nim* with two players and two piles of stones, at each turn a player removes some (non-zero) number of stones from one pile and the player who removes the last stone wins.

Claim. If the two piles contain the same number of stones at the start of the game, then the second player can always win.

Extra. What if the number of stones is not same in 2 piles?

Problem 8. Prove that for all integers $n \geq 1$ the following inequality holds:

$$\frac{1 \cdot 3 \cdot 5 \cdots (2n+1)}{2 \cdot 4 \cdot 6 \cdots (2n+2)} \geq \frac{1}{2n+2}.$$

Problem 9. Prove by induction that for all integers $n \geq 1$,

$$1^2 + 3^2 + 5^2 + \cdots + (2n-1)^2 = \frac{1}{3}\, n\,(4n^2 - 1)$$

Challenge Problems

Problem 10. Prove that the harmonic series diverges:

$$\sum_{n=1}^{\infty} \frac{1}{n} = \infty.$$

Problem 11. (**Partitions**) For every positive integer n, let $p(n)$ denote the number of ways to express n as a sum of positive integers. For instance,

$$p(4) = 5$$

because

$$4 = 4 = 3 + 1 = 2 + 2 = 2 + 1 + 1 = 1 + 1 + 1 + 1.$$

Also define $p(0) = 1$.

Prove that For every positive integer n,

$$p(n) - p(n-1)$$

is the number of ways to express n as a sum of integers each strictly greater than 1.

Problem 12. Mininim (remove 1,2 or 3; taking the last loses)
Consider the game of Mininim defined as follows: Some positive number of sticks are placed on the ground. Two players take turns removing one, two, or three sticks. The player to remove the last stick loses.

Statement: Use strong induction to show that the second player has a winning strategy if the number of sticks equals $4k + 1$ for some $k \in N$; otherwise, the first player has a winning strategy.